

Flavor profile of edible insects: grasshopper (*Schistocerca gregaria*) and honeybee brood (*Apis Mellifera*) as potential future food

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Introduction

Today, ca. 15% of the world's population suffers from protein and micronutrient deficits. This problem is expected to worsen in the near future as a result of the growing global population, which is expected to reach approximately 9 milliard by the year 2050.

Honey bees (*Apis mellifera*) have been cultivated by humans for thousands of years worldwide. The nutritional value of honey bee brood as an alternative source of nutrients contain 35% and 45% protein on a dry matter basis, respectively, similar to other protein sources such as poultry, beef and pork. Grasshopper (*Schistocerca gregaria*) is widely distributed in semi-arid and desert of Africa and reported to have more than 70% of proteins, and can be considered as an excellent source of nutrients and dietary components.

Drying is one of the first technological processing steps that affects nutritional value, as well as functional and sensory properties. Sensorial and nutritional information on edible insects is highly valuable in purpose to help consumers to overcome the "ick" factor towards insects and for producers to develop novel.



Grasshopper



Honey bee pupae

Materials and methods

Freeze dried honey bee (larvae/pupae, 1/1) and grasshopper (male and female, 1/1) were obtained from MIGAL Research Institute.

Other drying methods: oven, microwave.

Methods:

- Protein, fat content (AAOC,1990)
- Fatty acid composition (GC-FID)
- VOCs composition (GC-MS-SPME)
- Electronic nose
- Sensory evaluation – flavor profile analysis, intensity

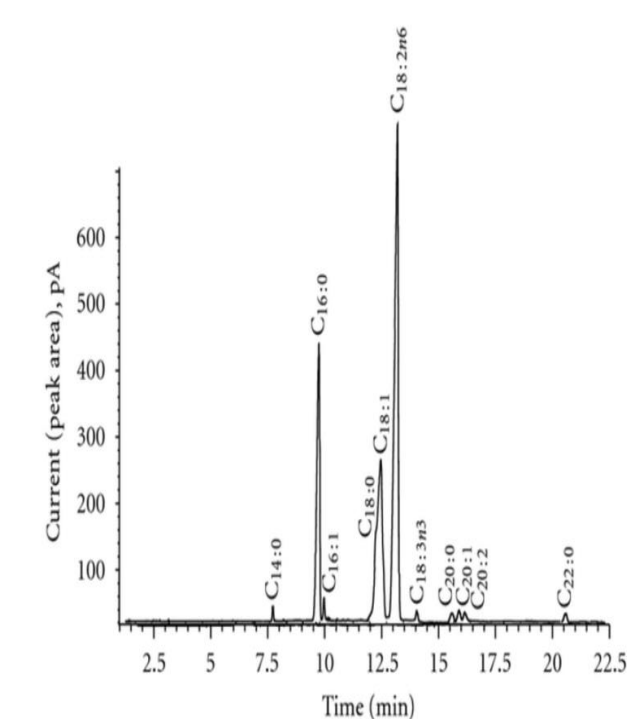


Fatty acid composition

Palmitic acid was the most highly detected SFA in the honey bee brood.

No significant difference between male and female grasshopper FA composition.

Fatty Acids	Larva (%)	Pupa (%)	Male (%)	Female (%)	*Cow milk powder(%)
Myristic Acid (C14:0)	3 ± 0.2	2.7 ± 0.3	1.6 ^a ± 0.2	1 ^b ± 0.12	12.7±0.3
Palmitic Acid (C16:0)	34.5 ^a ± 2.3	31.2 ^{ab} ± 2.7	20.1 ± 2.6 ^a	15.9 ± 1.4 ^b	33.9±0.9
Stearic Acid (C18:0)	10.4 ± 1.3	11.3 ± 1.5	11.5 ± 1.2	11.2 ± 0.9	8.5±0.52
Arachidic Acid (C20:0)	1 ± 0.1	1.8 ± 0.2	1.4 ± 0.07	1.7 ± 0.1	
Behenic Acid (C22:0)	1 ± 0.2	2 ± 0.2	1.1 ^a ± 0.1	1.7 ^b ± 0.01	
Saturated	50 ± 4.23	49.1 ± 4.9	35.7 ± 4.2	31.5 ± 2.5	67.6
Oleic Acid (C18:1)	45.9 ± 5.3	46.6 ± 6.3	25.3 ± 3.2	24.1 ± 1.9	25.4±0.2
Linoleic Acid (C18:2)	1.5 ± 0.2	2.1 ± 0.1	18 ± 2	21.1 ± 2.5	1.11±0.01
Linolenic Acid (C18:3)	2.6 ± 0.4	2.2 ± 0.2	21.1 ± 2.6	23.4 ± 2.7	0.72±0.03
Unsaturated	50 ± 5.9	50.9 ± 6.6	64.3 ± 7.9	68.5 ± 9.7	32.4



Results and discussion

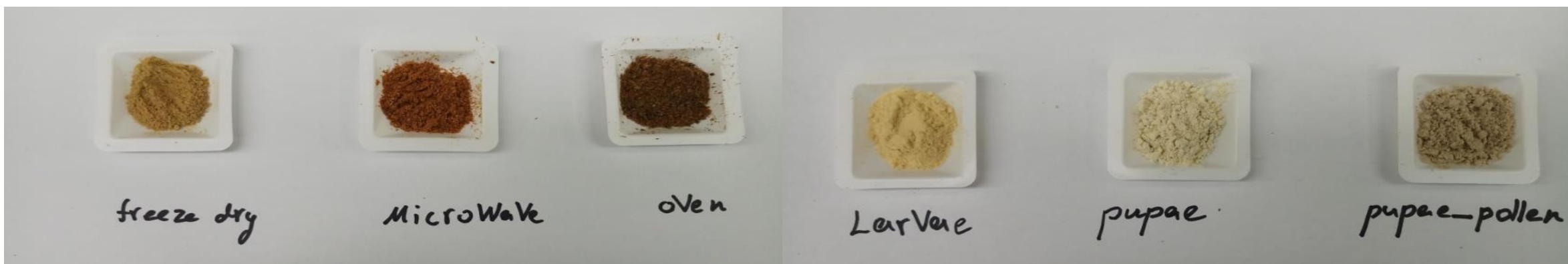
Nutritional value of honey bee brood and grasshopper

	Protein(%)	Fat(%)	Ash(%)
Larvae	19 ± 0.3 ^b	28.1 ± 0.3 ^a	2.8 ± 0.2
Pupae	24.6 ± 0.2 ^a	21.1 ± 1.1 ^b	3.2 ± 0.1
Male	52.9 ± 2.6 ^a	13.1 ± 0.6 ^a	2.9 ± 0.1 ^a
Female	54 ± 0.9 ^a	8.5 ± 0.1 ^b	3 ± 0.1 ^a

Protein value increased significantly with brood age.

Amino Acid	Larva (%)	Pupa(%)	Grasshopper (%)	*Egg protein (%)
Leu	8.4 ± 0.1	8.25	7.5 ± 0.1	9.9
Thr	2.8 ± 0.1	2.9 ± 0.3	3 ± 0.1	6.1
Phe	3.2 ± 0.1	3.3	2.7 ± 0.1	3.9
Lys	7.2 ± 0.3	7.3	5.3 ± 0.3	7.8
Ile	5.6 ± 0.1	5.5	4.5	5.2
Val	7.9 ± 0.1	7.8	8.1	6.1
Met	3.3 ± 0.1	3.22	1.9	3.5
Essential	38.4 ± 0.6	38.3 ± 0.3	35 ± 0.2	42.6

Drying methods – Grasshopper



Freeze Dry: -51°, 0.01bar
Microwave- Medium heat, 20 minutes
Oven: 55°C 48 hour

Honeybee brood

Freeze Dry: -51°, 0.01bar

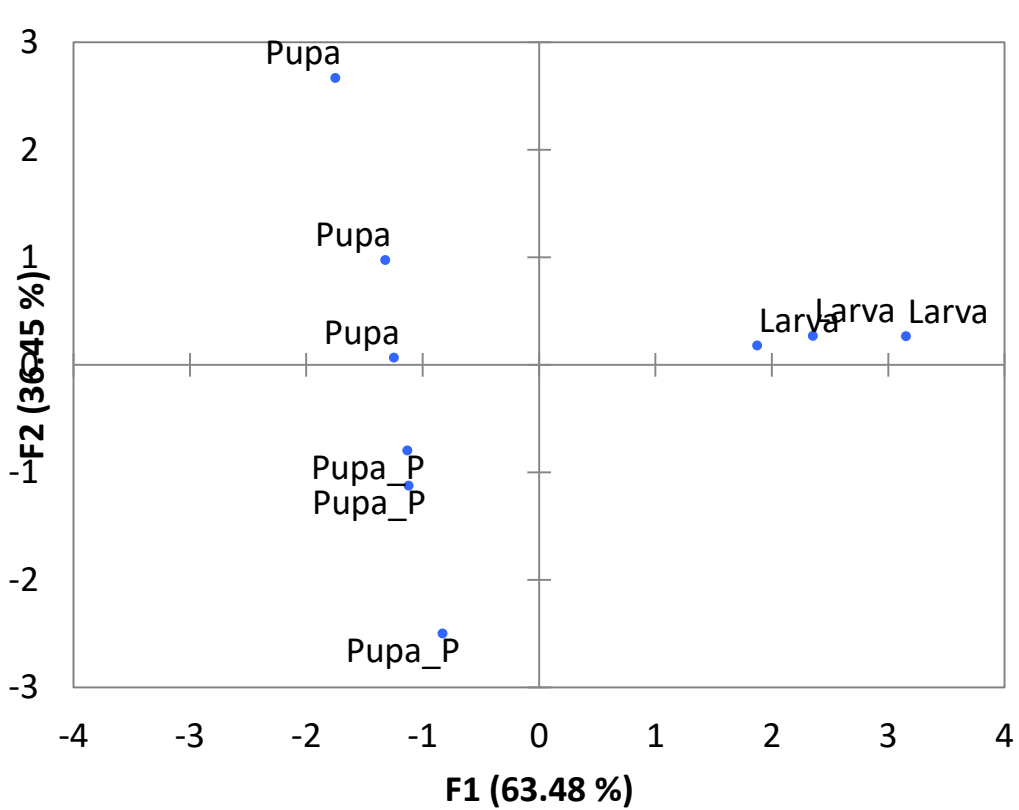
Gel hardness increased as a function of pH value with the maximum at pH 7 and 9 ($p \leq 0.01$)

Electronic tongue and nose

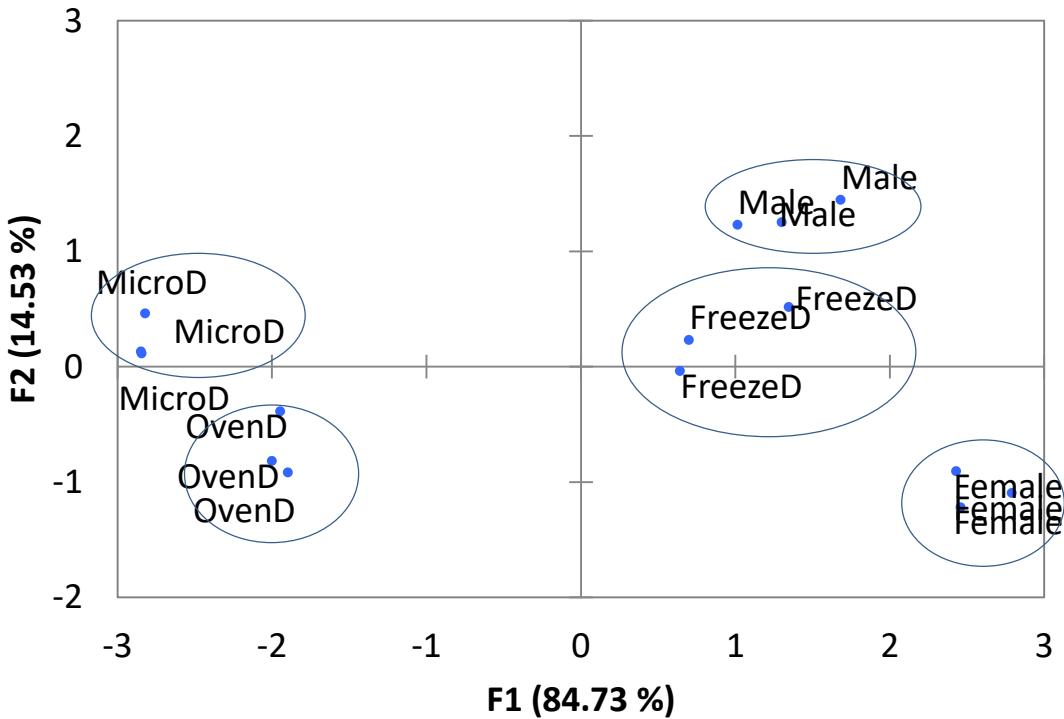


Flavor profile

E-nose PCA plot – honeybee brood

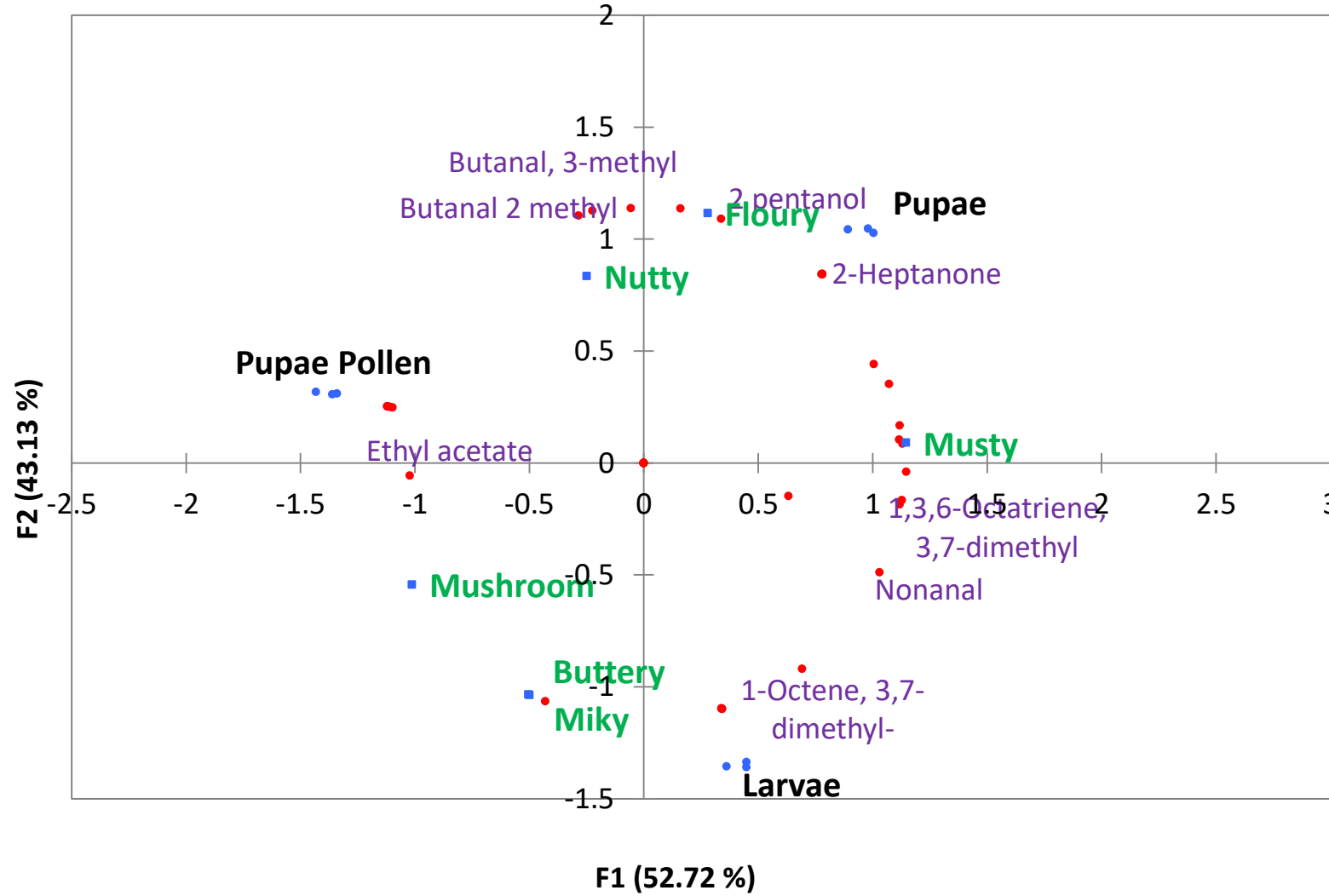


E-nose – Grasshopper



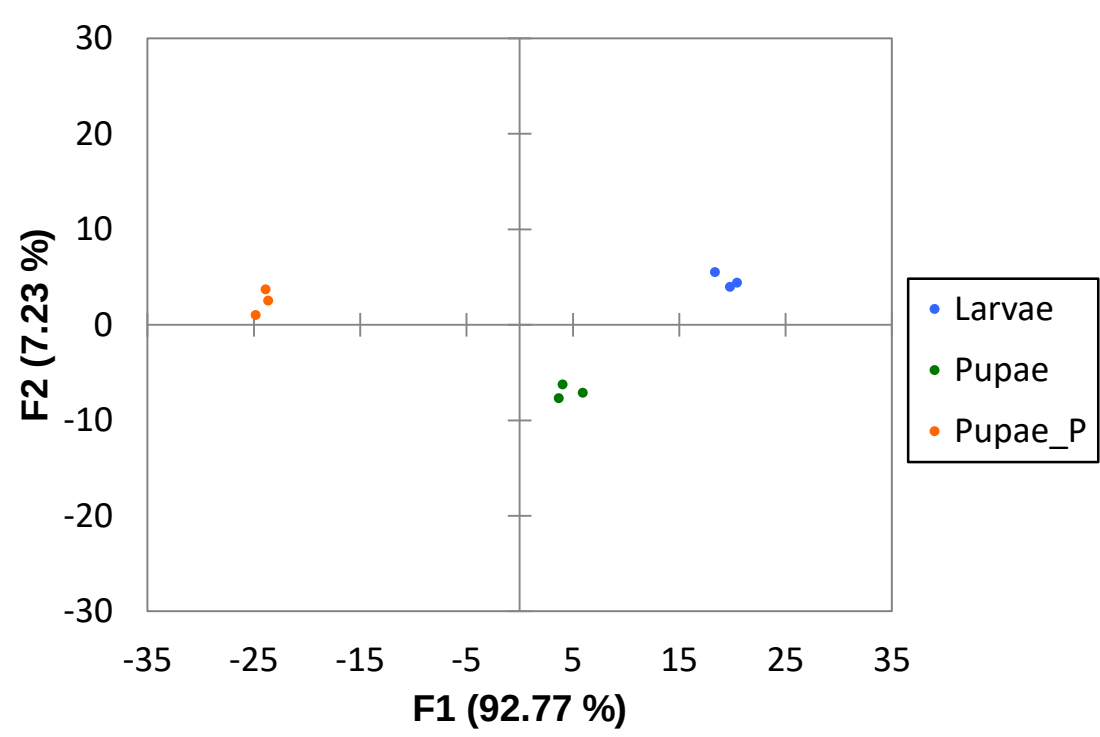
Volatile analysis – GC-MS

Honeybee brood

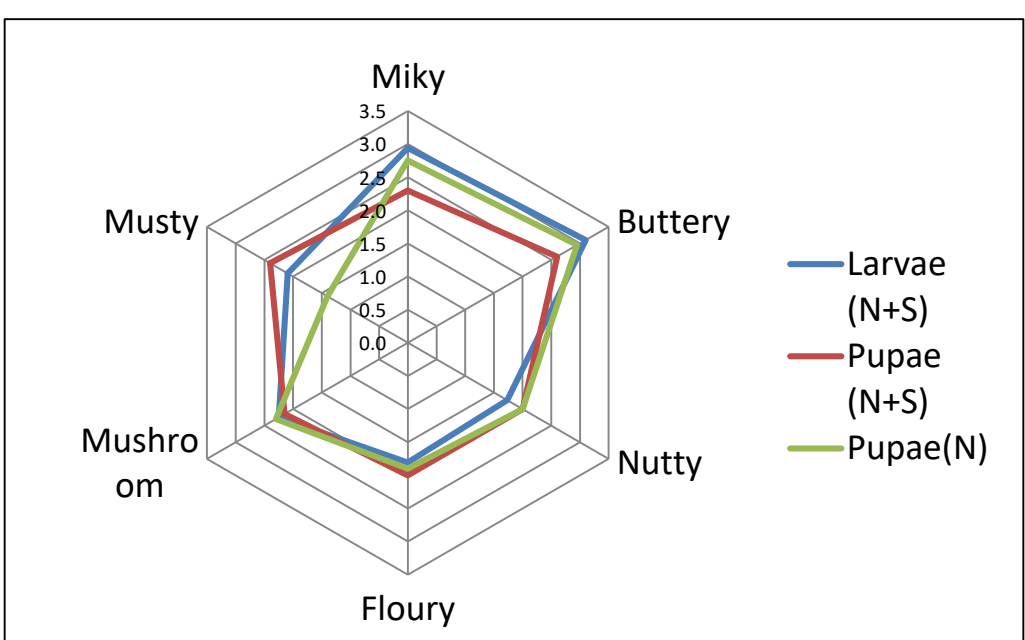


- Most of volatiles originated from larvae and pupae were odorless alkanes and alkenes
- The difference in volatile profiles between the developmental stages may derive from the variation in the insect's hydrocarbon composition
- In larvae, the main odor compounds were diacetyl and nonanal with a fatty and buttery fragrance, ocimene with a green-flowery odor, and dimethyl sulfide with a sulfurous odor

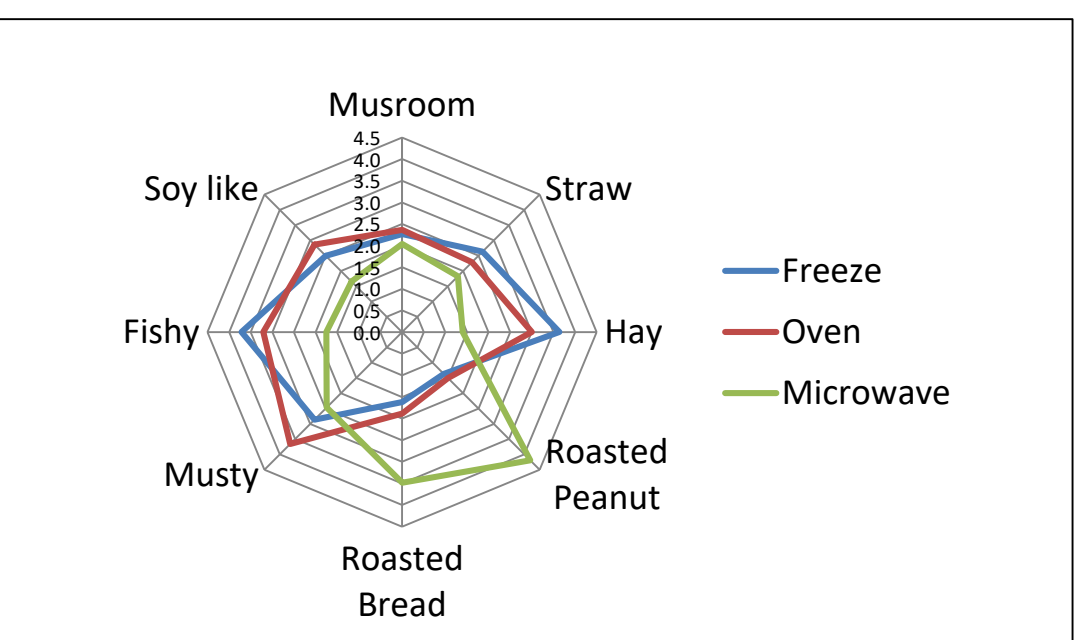
Electronic tongue – Honeybee brood



Sensory evaluation



Honey bee N+S larvae and N+S pupae received the same descriptors: milky, buttery, nutty, floury, musty and mushroom-like.



Oven and freeze dry methods induced the appearance of distinct "fishy" notes, which can be related to protein and lipid oxidation



Conclusion

- The odor generated from honey bee brood powders was comprehensively evaluated for honey bees at different life stages and reared on different diets
- The composition and intensity of volatile compounds were significantly affected by the studied drying method.
- Drying in a microwave led to additional emission of pyrazine and pyrrole compounds and the emergence of a "roasted" odor that is typical for the Maillard reaction
- choice of drying method can serve to modify the volatile composition and odor characteristics of edible insects
- Our result may support the industry in developing novel food products based on edible insects such as honey bees by matching the preferred odors to the food.

Acknowledgements

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